Table S1: Inclusion, exclusion criteria, study design and outcome assessments of the included studies

Trails	**Kappos et al 2006**
Inclusion Criteria	Patients aged 18y to 60y with RRMS and at least one of the following: two or more documented relapses during the previous 2 years, one or more documented relapses in the year before enrollment, and one or more gadolinium-enhanced lesions detected on MRI at screening. Additional eligibility criteria included a score of 0 to 6 on the EDSS and with no evidence of relapse for at least 30 days before screening and during the screening and baseline phases.
Exclusion Criteria	Previously use of corticosteroids (within 30 days), immunomodulatory therapy (within 3 months), or immunosuppressive treatment (e.g., azathioprine or methotrexate within 6 months, cyclophosphamide within 12 months, or mitoxantrone or cladribine within 24 months); a history of cardiac conditions that might increase the risk of a decrease in heart rate; a white-cell count of less than 3500 per cubic millimeter; and a lymphocyte count of less than 800 per cubic millimeter.
Study Design	Patients were randomly assigned, in a 1:1:1 ratio, to 1.25 mg of fingolimod, 5.0 mg of fingolimod, or a matching placebo once daily; all drugs were given as capsules.
Efficacy Outcomes	The total number of gadolinium-enhanced lesions per patient recorded on T1-weighted MRI at monthly intervals for 6 months; the total volume of gadolinium-enhanced lesions per patient; the proportion of patients with gadolinium-enhanced lesions; the total number of new lesions per patient on T2-weighted images; changes in lesion volume on T2-weighted images; the brain volume from baseline to month 6; the number of patients remaining free of relapse; the ARR; the time to the first relapse; the Hamburg Quality of Life Questionnaire in Multiple Sclerosis score and the BDI score.
Safety Outcomes	Adverse events and serious adverse events.

Trails	**Cohen et al 2010 (TRANSFORMS)**
Inclusion Criteria	Patients aged 18y to 55y with RRMS and had at least one documented relapse during the previous year or at least two documented relapses during the previous 2 years, and had a score of 0 to 5.5 on the EDSS.
Exclusion Criteria	A documented relapse or corticosteroid treatment within 30 days before randomization, active infection, macular edema, immunosuppression (either drug- or disease-induced), and clinically significant coexisting systemic disease.
Study Design	Patients were randomly assigned to 12 months of treatment with oral fingolimod, at a daily dose of either 1.25 or 0.5 mg, or intramuscular interferon beta-1a, at a weekly dose of 30 μg.
Efficacy Outcomes	The ARR; the number of new or enlarged hyperintense lesions on T2-weighted MRI scans at 12 months and the time to confirmed disability progression.
Safety Outcomes	Adverse events and death.

Trails	Kappos et al 2010 (FREEDOMS)
Inclusion Criteria	Patients aged 18y to 55y with RRMS and had one or more documented relapses in the previous year or two or more in the previous 2 years; and a score of 0 to 5.5 on the EDSS.
Exclusion Criteria	Relapse or corticosteroid treatment within 30 days before randomization, active infection, macular edema, diabetes mellitus, immune suppression (drug- or disease-induced), or clinically significant systemic disease.
Study Design	Patients were randomly assigned, in a 1:1:1 ratio, to receive oral fingolimod capsules in a dose of 0.5 mg or 1.25 mg or matching placebo, once daily for 24 months.
Efficacy Outcomes	The ARR; the time to confirmed disability progression; the time to a first relapse; time to disability progression (confirmed after 6 months); changes in the EDSS score and MSFC z score between baseline and 24 months; number of gadolinium-enhancing lesions; proportion of patients free from gadolinium-enhancing lesions; number of new or enlarged lesions on T2-weighted MRI scans; proportion of patients free from new or enlarged lesions on T2-weighted scans; volumes of hyperintense lesions on T2-weighted scans and hypointense lesions on T1-weighted scans; change in brain volume between baseline and 24 months.
Safety Outcomes	Adverse events and death.

Trails	**Saida et al 2012**
Inclusion Criteria	Patients aged 18y to 60y with RRMS and had one or more documented relapses in the previous year or two or more in the previous 2 years, or at least one gadolinium (Gd)-enhanced T1-weighted brain lesion within the 30 days prior to study commencement; had to have at least one T2-weighted brain lesion and a score of 0–6 on the EDSS.
Exclusion Criteria	Patients with long cord lesions of at least three vertebral segments on spinal MRI at or prior to screening; primary progressive MS, relapse or corticosteroid treatment within 30 days before randomization, malignancy, macular oedema, diabetes mellitus, active infection, immune suppression (drug or disease induced), clinically significant systemic disease, or pregnancy; received cladribine, cyclophosphamide, mitoxantrone, or other immunosuppressive or immunoglobulin medication in the six months prior to randomization; had plasmapheresis immunoadsorption or interferon beta therapy in the three months prior to randomization.
Study Design	Patients were randomly assigned, in a 1:1:1 ratio, to receive once-daily fingolimod capsules, 0.5 mg or 1.25 mg, or matching placebo for six months.
Efficacy Outcomes	The percentage of patients free from Gd-enhanced lesions at both months 3 and 6; the percentage of patients free from relapses over six months; the number of Gd-enhanced lesions at six months; the number of new/enlarging T2 lesions over six months; the proportions of patients free from new/enlarging T2 lesions over six months; the proportions of patients free from new MRI activity (Gd-enhanced lesions and new/enlarging T2 lesions); the ARR.
Safety Outcomes	Adverse events and serious adverse events.

Trails	**Calabresi et al 2014 (FREEDOMS II)**
Inclusion Criteria	Patients aged 18y to 55y with RRMS and had one or more confirmed relapses during the preceding year (or two or more confirmed relapses during the previous 2 years), had an EDSS 10 score of 0–5.5, and had no relapse or steroid treatment within 30 days before randomization.
Exclusion Criteria	Patients with clinically significant systemic disease or immune suppression (drug-induced or disease induced), active infection or macular oedema, diabetes mellitus, or a history of malignancy (apart from successfully treated basal or squamous-cell skin carcinoma), and patients with specific cardiac, pulmonary, or hepatic disorders.
Study Design	A 24-month, randomized, double-blind, placebo-controlled, parallel-group, multicentre study comparing the efficacy of once-daily fingolimod (0.5 mg and 1.25 mg doses) with placebo.
Efficacy Outcomes	The ARR; percent brain-volume change from baseline and time to disability progression; the time to first relapse and proportion of relapse free patients; time to disability progression confirmed at 6 months, as measured by EDSS; change from baseline to the end of study on the MSFC score; and effect on MRI measurements of inflammatory disease activity, and MRI measurements of burden of disease.
Safety Outcomes	Adverse events and death.

Trails	Fox et al 2014 (EPOC)
Inclusion Criteria	Patients had been treated with an DMT for at least 6 months before screening and entered the study without an intervening washout period; adults under 65 years of age with RRMS; an EDSS score of 0–5.5, and treatment-naïve to fingolimod.
Exclusion Criteria	History of chronic disease of the immune system (except MS), history of active malignancy (except localized basal/ squamous cell carcinoma of the skin), uncontrolled diabetes mellitus (glycosylated hemoglobin > 7%), macular edema present at screening, active systemic bacterial, viral, or fungal infection (s), pregnancy, and certain cardiovascular, pulmonary, and hepatic conditions.
Study Design	A 6-month, randomized, open-label, multicenter trial with an optional 3-month extension, patients were randomized using an interactive voice response system to either the once-daily fingolimod 0.5mg arm or the DMT arm,in a 3:1 ratio.
Efficacy Outcomes	Changes in treatment satisfaction between baseline and month 6 measured by TSQM; changes in fatigue severity scale, depression (BDI), activities of daily living, and health-related QOL (SF-36) between baseline and month 6; the seven-point clinical global impression of improvement scale.
Safety Outcomes	Adverse events and death.

Trails	**Comi et al 2017 (GOLDEN)**
Inclusion Criteria	Patients aged 18 to 60 years and diagnosed with RRMS with active disease (at least one clinical relapse in the past year, or two clinical relapses in the past 2 years if there were signs of disease activity in one brain MRI scan performed in the past 6 months) and cognitive impairment at screening.
Exclusion Criteria	Unsatisfactory response with multi-weekly IFNs (IFN β-1a/b); hyperactive forms of MS; EDSS score > 5.0; acute MS relapse < 30 days before screening; prior or current diagnosis of major depression according to the Diagnostic and Statistical Manual of Mental Disorders-Text Revision and history of any chronic disease of the immune system other than MS.
Study Design	An 18-month multicentre, open-label, rater-blinded, randomised, parallel-group pilot study conducted in patients with RRMS.
Efficacy Outcomes	Cognitive impairment; executive functions; depression score; MRI parameters (T2-hyperintense, T1-hypointense and T1-enhancing lesions, normalised brain volume and percentage brain volume change versus screening scan); MS relapses; EDSS scores.
Safety Outcomes	Adverse events and serious adverse events.

Trails	**Cree et al 2018 (PREFERMS)**
Inclusion Criteria	Patients aged 18 – 65 years, diagnosed with RRMS and with an EDSS score up to 6; Patients naïve to treatment or who have been treated with no more than one class of DMT previously.
Exclusion Criteria	Use of other investigational drugs within 30 days of screening; history of hypersensitivity to any of the study drugs or to drugs of similar chemical classes; prior exposure to fingolimod or any other sphingosine-1-phosphate-receptor-modulating compounds; history or presence of malignancy of any organ system; patients diagnosed with secondary progressive MS or primary progressive MS; patients with a history of chronic disease of the immune system other than MS or a known immunodeficiency syndrome; patients with uncontrolled diabetes mellitus (glycated hemoglobin >7%); diagnosis of macular edema during the screening phase; patients with active systemic bacterial, viral, or fungal infections; patients without a history of chickenpox or without vaccination against varicella-zoster virus at screening; patients who have received any live or live attenuated vaccines; patients with any medically unstable condition as assessed by the investigator.
Study Design	A randomized, open-label, active-controlled, parallel-group, multicenter study that followed up patients for 48 weeks, patients were randomized 1:1 to fingolimod 0.5mg/day or to an DMT using an interactive voice-and-web-response system.
Efficacy Outcomes	Patient retention on randomized treatment over 48 weeks; reasons for discontinuation of randomized treatment; cognitive impairment (SDMT); changes in brain volume (measured as percentage change from baseline using MRI); and patient-reported satisfaction; the ARR; cumulative number of newly active gadolinium-enhanced (Gd+) T1 lesions; number of new/enlarged T2 lesions; number of new active lesions; change in Gd+ lesion count; and changes in cortical graymatter and thalamic volume measures.
Safety Outcomes	Adverse events and serious adverse events.

Trails	**Cree et al 2020 (ASSESS)**
Inclusion Criteria	Eligible participants were between ages 18 and 65 years, had received a diagnosis of RRMS, had experienced at least 1 documented relapse during the previous year or 2 documented relapses during the previous 2 years before randomization, had received an EDSS score between 0 and 6.0 points (both inclusive) at screening, and had not experienced a relapse within 30 days of randomization.
Exclusion Criteria	Patients with a history of malignancy other than cutaneous basal cell carcinoma in the last 5 years, active infection, diabetes mellitus, macular edema or clinically significant systemic disease, active chronic diseases of the immune system other than MS, severe hepatic injury or other hepatic conditions, certain electrocardiogram findings, or history of certain cardiovascular conditions within 6 months were excluded. Patients were also excluded if they received the following treatments prior to randomization: immunosuppressive/chemotherapeutic medications within 6 months; immunoglobulins within 4 weeks; natalizumab within 2 months; rituximab, alemtuzumab, ofatumumab, ocrelizumab, or cladribine within one year, or teriflunomide within 3.5 months.
Study Design	A phase 3b multicenter randomized parallel-group rater-blinded and dose-blinded 12-month clinical trial, eligible adult participants were randomized (1:1:1) to receive fingolimod, 0.5mg, or fingolimod, 0.25 mg, orally once per day or glatiramer acetate, 20 mg, subcutaneously once per day using an interactive voice response system.
Efficacy Outcomes	The ARR; the number of new or newly enlargingT2 lesions, the number of gadolinium-enhancing T1 lesions and the proportion of participants free from these lesions, the volume of gadolinium-enhancing T1 lesions, the change from baseline in T2 lesion volume, the T1-hypointense lesion volume and brain volume, the change from baseline in the TSQM; scores from the MSFC; the SDMT score; the Patient-Reported Indices in Multiple Sclerosis-Activities scale; the 29-item Multiple Sclerosis Impact Scale; the proportion of relapse-free participants, the time to first relapse, and the severity of relapses.
Safety Outcomes	Adverse events and serious adverse events.

Trails	**NCT01534182** **(EPOC)**
Inclusion Criteria	Patients must be diagnosed with RRMS; Patients who explicitly agree to be assigned to a treatment group that may receive or DMT after having been informed about their respective benefits and possible adverse events by the investigator; Male or female patients aged 18-70 years; An EDSS score of 0-6 inclusive; Must have received continuous treatment with a single approved and indicated MS DMT for a minimum of 6 months prior to the screening visit; Patients must continue with this MS DMT until the randomization visit; Naïve to treatment with fingolimod.
Exclusion Criteria	A manifestation of MS other than those defined in the inclusion criteria; A history of chronic disease of the immune system other than MS or a known immunodeficiency syndrome; History of malignancy of any organ system; Diagnosis of macular edema during Screening Phase; Patients with active systemic bacterial, viral or fungal infections, or known to have AIDS or to have positive HIV antibody test; Patients who have received any live or live attenuated vaccines (including for varicella-zoster virus or measles) within 2 months prior to baseline; Patients who have received total lymphoid irradiation or bone marrow transplantation; History of selected immune system treatments and/or medications; Any medically unstable condition, as assessed by the investigator; Selected cardiovascular, or hepatic conditions; Selected abnormal laboratory values; Patients with any other disease or clinical condition (including neurologic or psychiatric disorders) which may affect patient enrollment into the study and study medication use by the Investigators' opinion; Participation in any clinical research study evaluating another not approved in Russia investigational drug or therapy within 6 months prior to baseline; History of hypersensitivity to the study drug or to drugs of similar chemical classes; Pregnant or nursing (lactating) women.
Study Design	A 6-month, randomized, active comparator, open-label, multi-center study to evaluate patient outcomes, safety and tolerability of fingolimod 0.5 mg/day in patients with relapsing remitting multiple sclerosis who are candidates for multiple sclerosis therapy change from previous DMT.
Efficacy Outcomes	Change in patient-reported treatment satisfaction (TSQM); changes in patient-reported effectiveness, side effects and convenience; change in patient-reported depression (BDI); change in patient-reported health-related quality-of-life using the SF-36.
Safety Outcomes	Adverse events, serious adverse events and death.

Trails	**NCT01317004** **(EPOC)**
Inclusion Criteria	Patients must be diagnosed with RRMS; Patients who explicitly agree to be assigned to a treatment group that may receive or DMT after having been informed about their respective benefits and possible adverse events by the investigator; Male or female patients aged 18-65 years; An EDSS score of 0-5.5 inclusive; Must have received continuous treatment with a single approved and indicated MS DMT for a minimum of 6 months prior to the screening visit; Patients must continue with this MS DMT until the randomization visit; Naïve to treatment with fingolimod.
Exclusion Criteria	A manifestation of MS other than those defined in the inclusion criteria; A history of chronic disease of the immune system other than MS or a known immunodeficiency syndrome; History of malignancy of any organ system (other than localized basal cell carcinoma of the skin), treated or untreated, within the past 5 years, regardless of whether there is evidence of local recurrence or metastases; Patients with uncontrolled diabetes mellitus (HbA1c > 7%); Diagnosis of macular edema during Screening Phase.
Study Design	A 6-month, randomized, active comparator, open-label, multi-center study to evaluate patient outcomes, safety and tolerability of fingolimod (FTY720) 0.5 mg/day in patients with relapsing remitting multiple sclerosis who are candidates for MS therapy change from previous DMT.
Efficacy Outcomes	Change in patient-reported treatment satisfaction (TSQM); change from baseline in patient-reported activities of daily living; change from baseline in patient-reported fatigue; changes in patient-reported effectiveness, side effects and convenience; change in patient-reported depression (BDI); change in patient-reported health-related quality-of-life using the SF-36; physician-reported clinical global impression of improvement.
Safety Outcomes	Adverse events, serious adverse events and death.

EDSS: Expanded Disability Status Scale; MRI: Magnetic Resonance Imaging; BDI: Beck Depression Inventory; RRMS: Relapsing-Remitting Multiple Sclerosis; ARR: Annualized Relapse Rate; MSFC: Multiple Sclerosis Functional Composite; DMT: Disease Modifying Therapy, including interferon beta 1a and 1b (IFNβ-1a and 1b) and glatiramer acetate (GA); TSQM: Treatment Satisfaction Questionnaire for Medication; SF-36: 36-item Short-Form Health Survey; SDMT: Symbol Digit Modalities Test.

Fig. S1 (ARR)

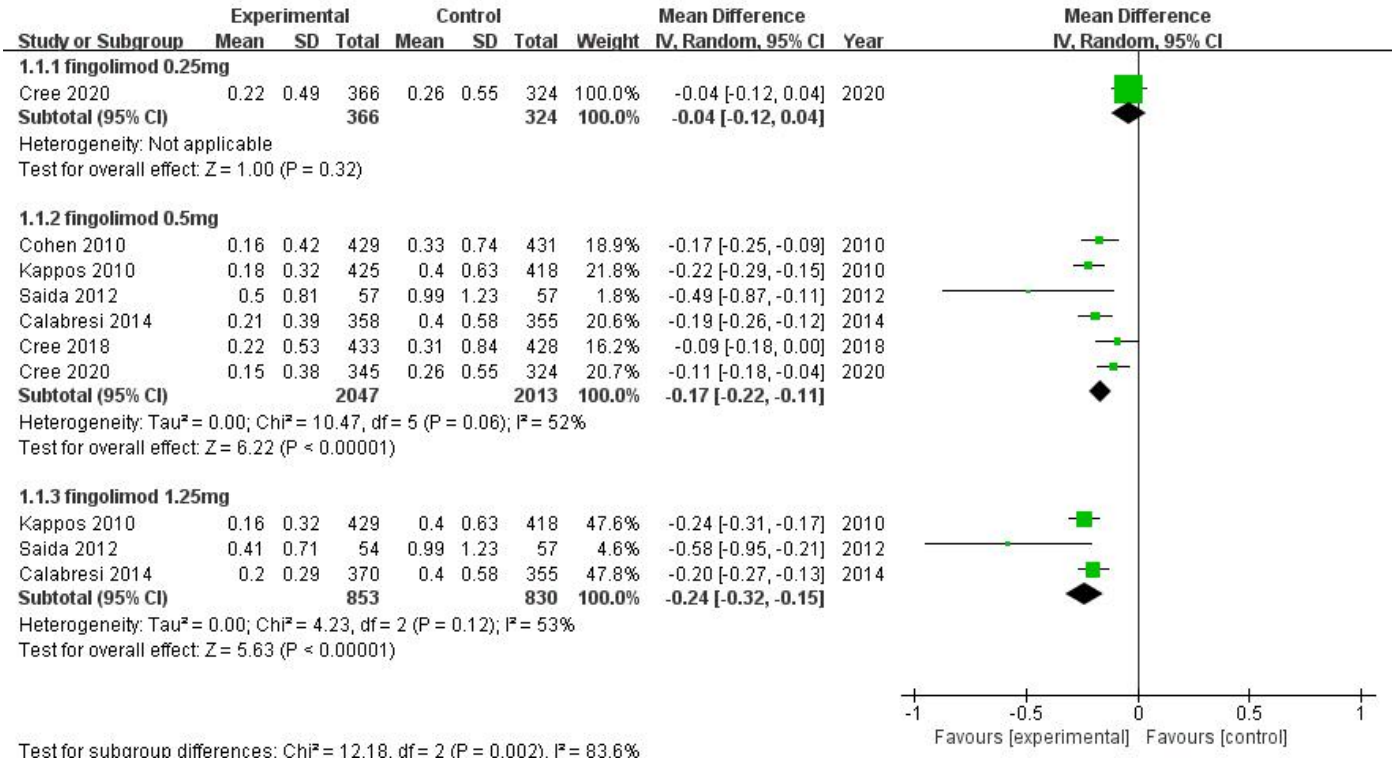


Fig. S2 (Number of patients free of relapse)

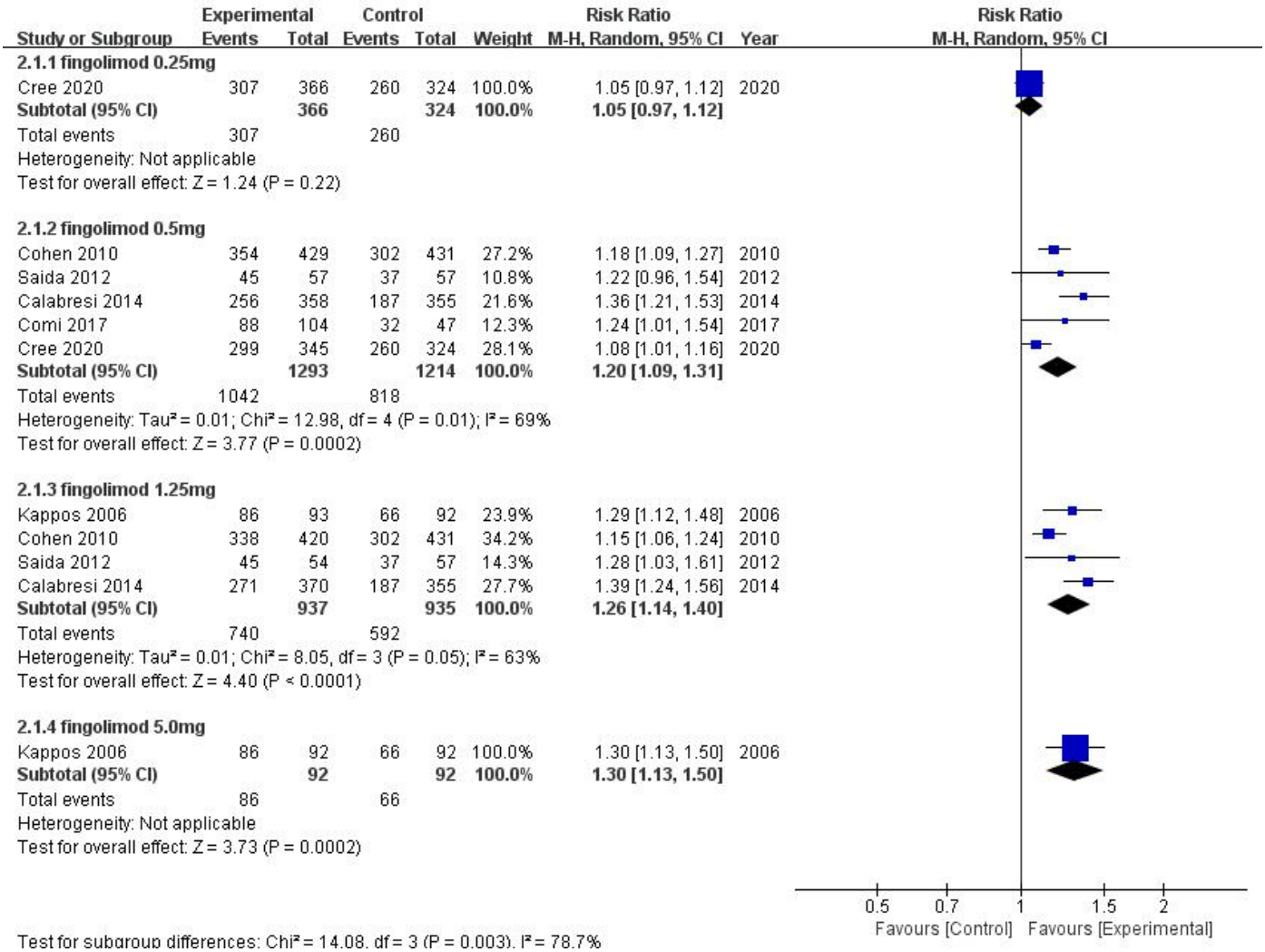


Fig. S3 (ΔEDSS)

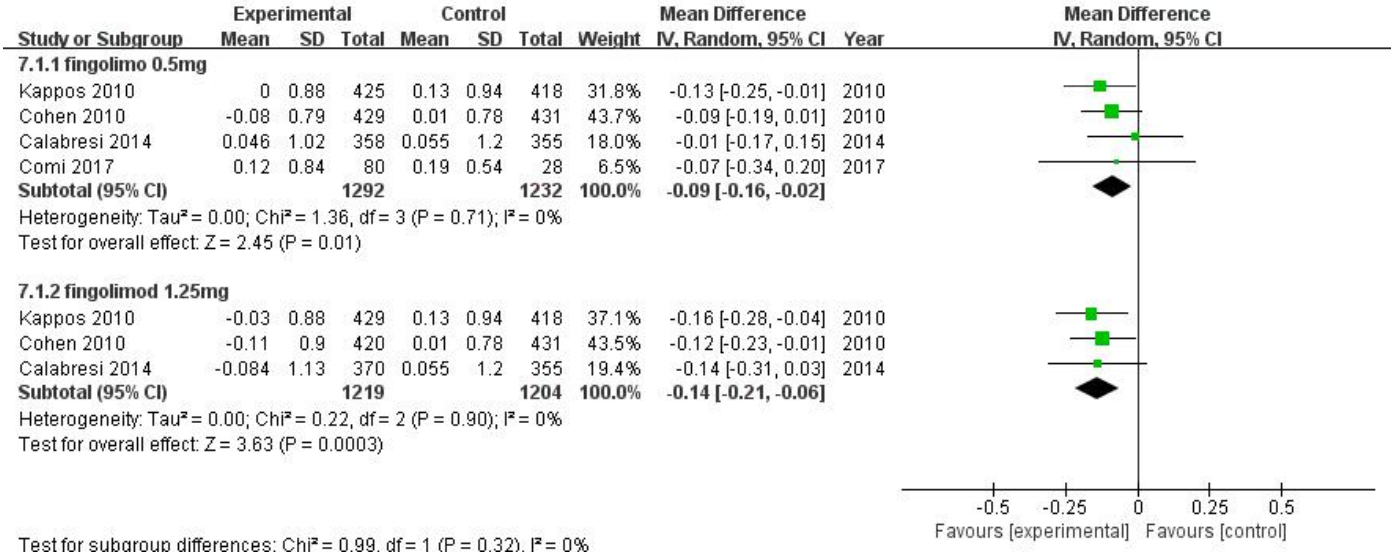


Fig. S4 (Number of patients free of increased gadolinium-enhanced lesions in T1)

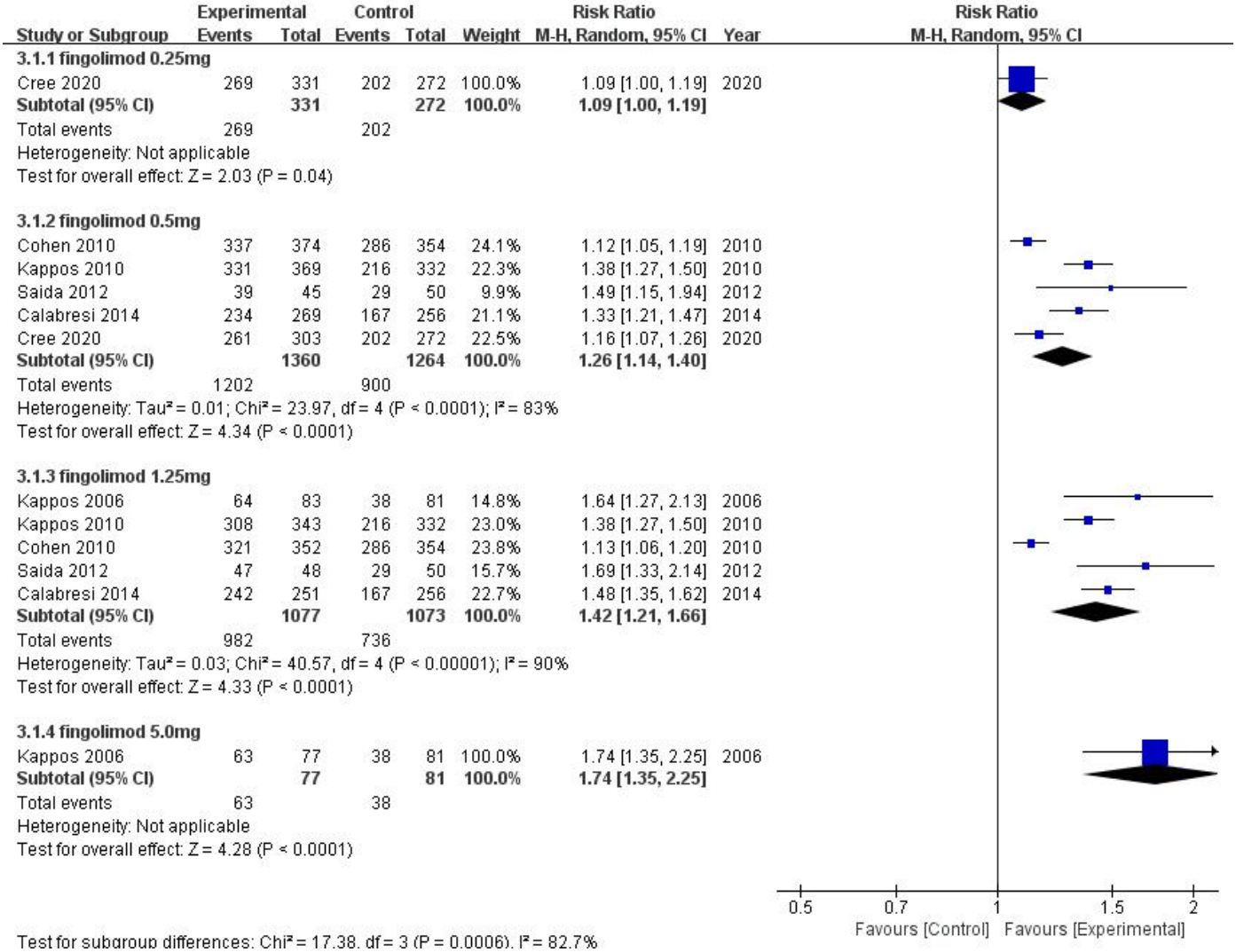


Fig. S5 (Number of patients with no new or newly enlarged lesions in T2)

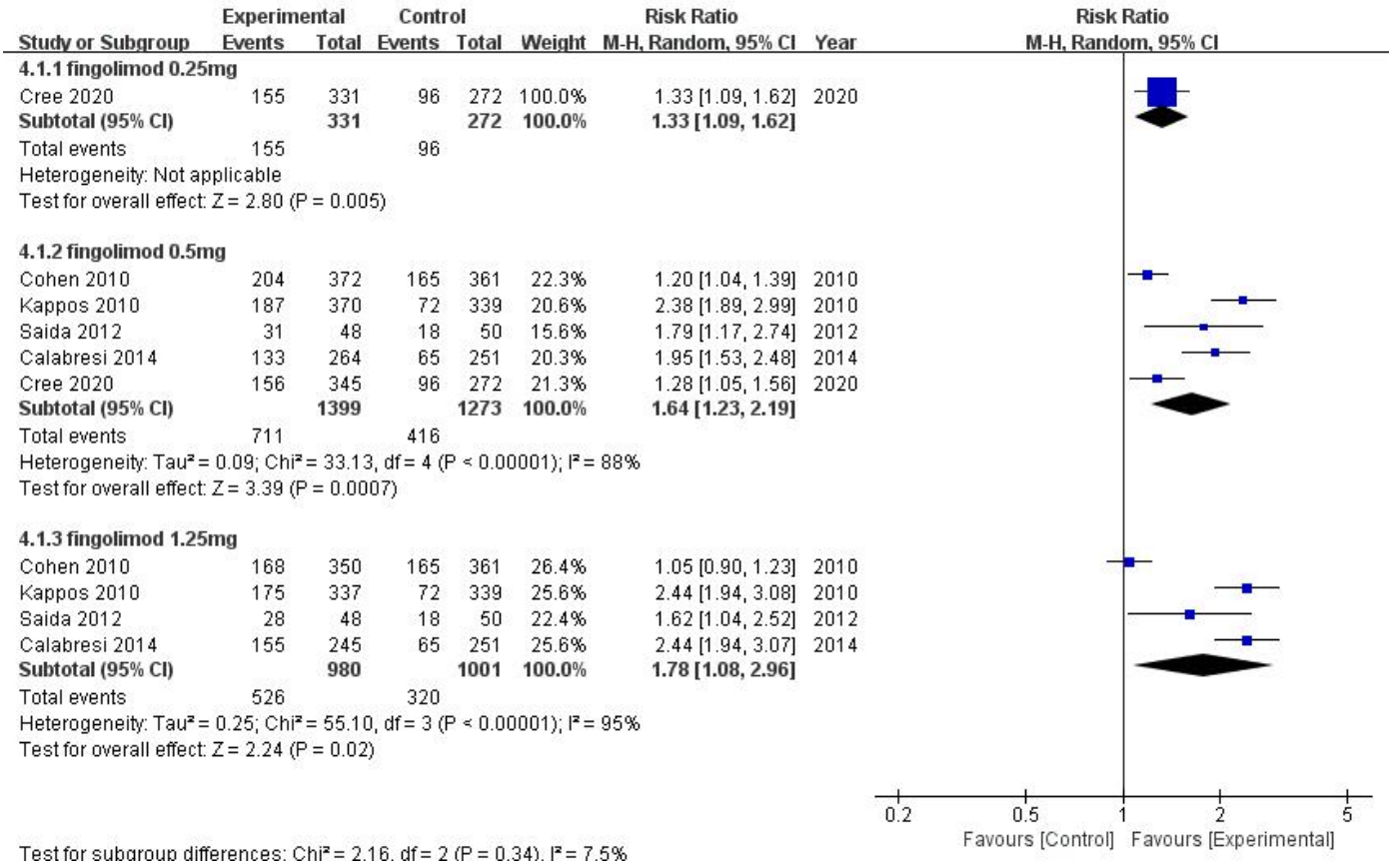


Fig. S6 (PBVC)

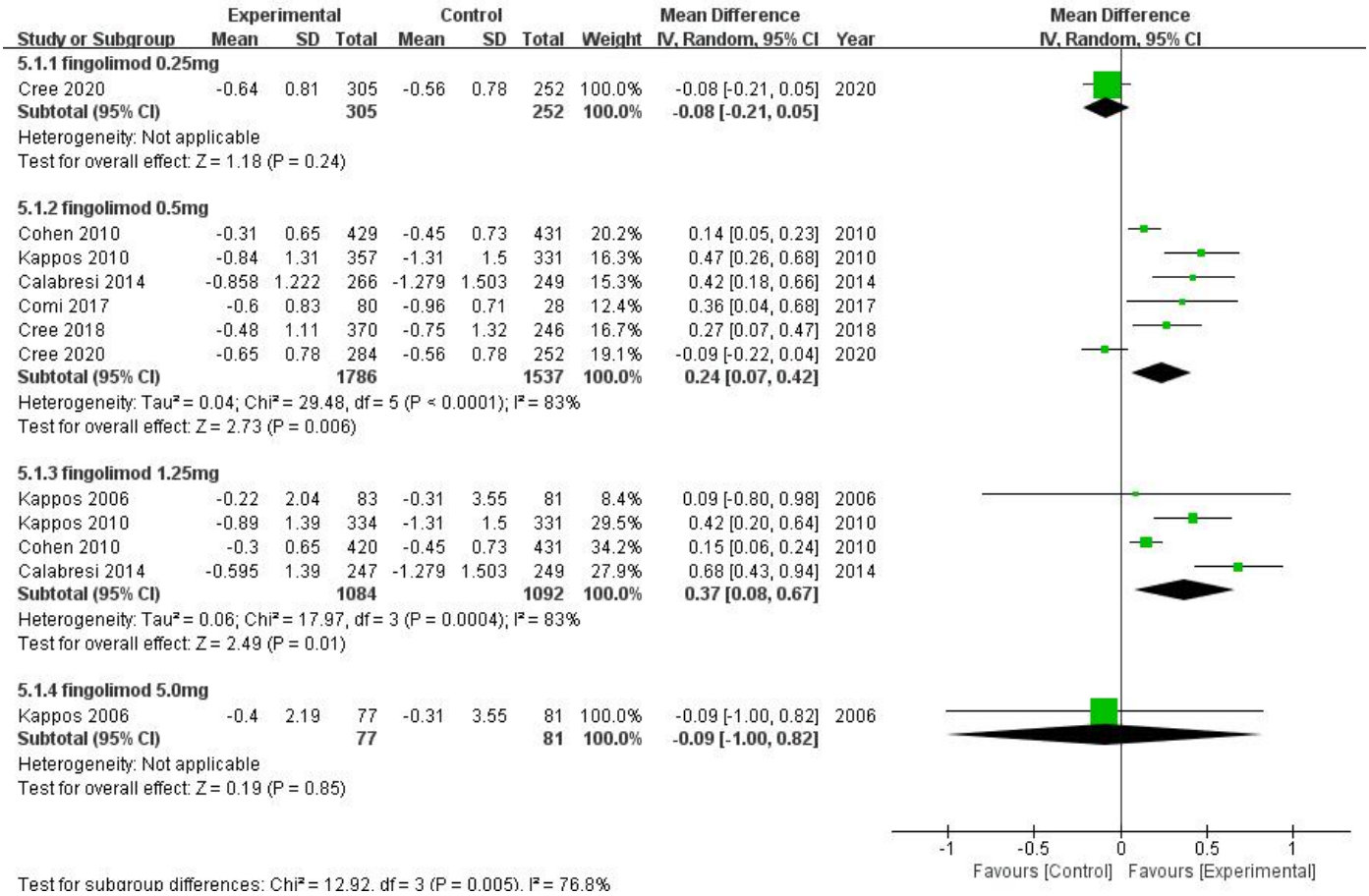


Fig. S7 (ΔBDI)

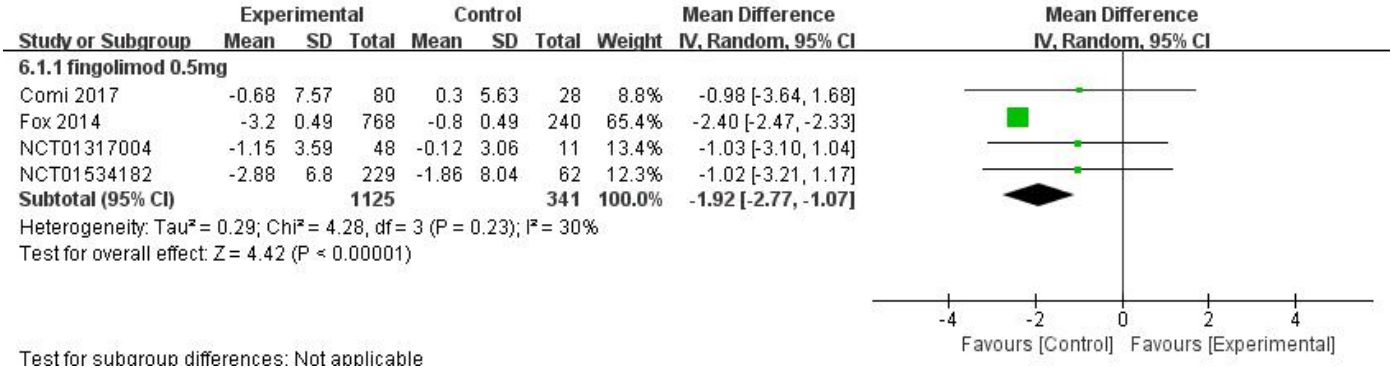


Fig. S8 (ΔTSQM)

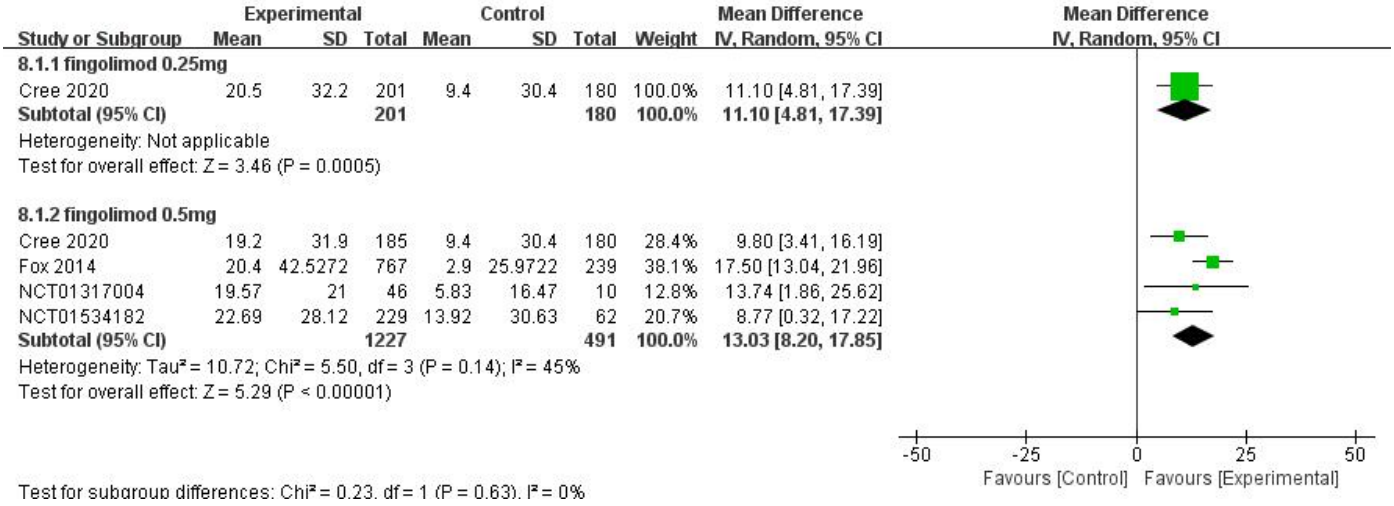


Fig. S9 (AEs)

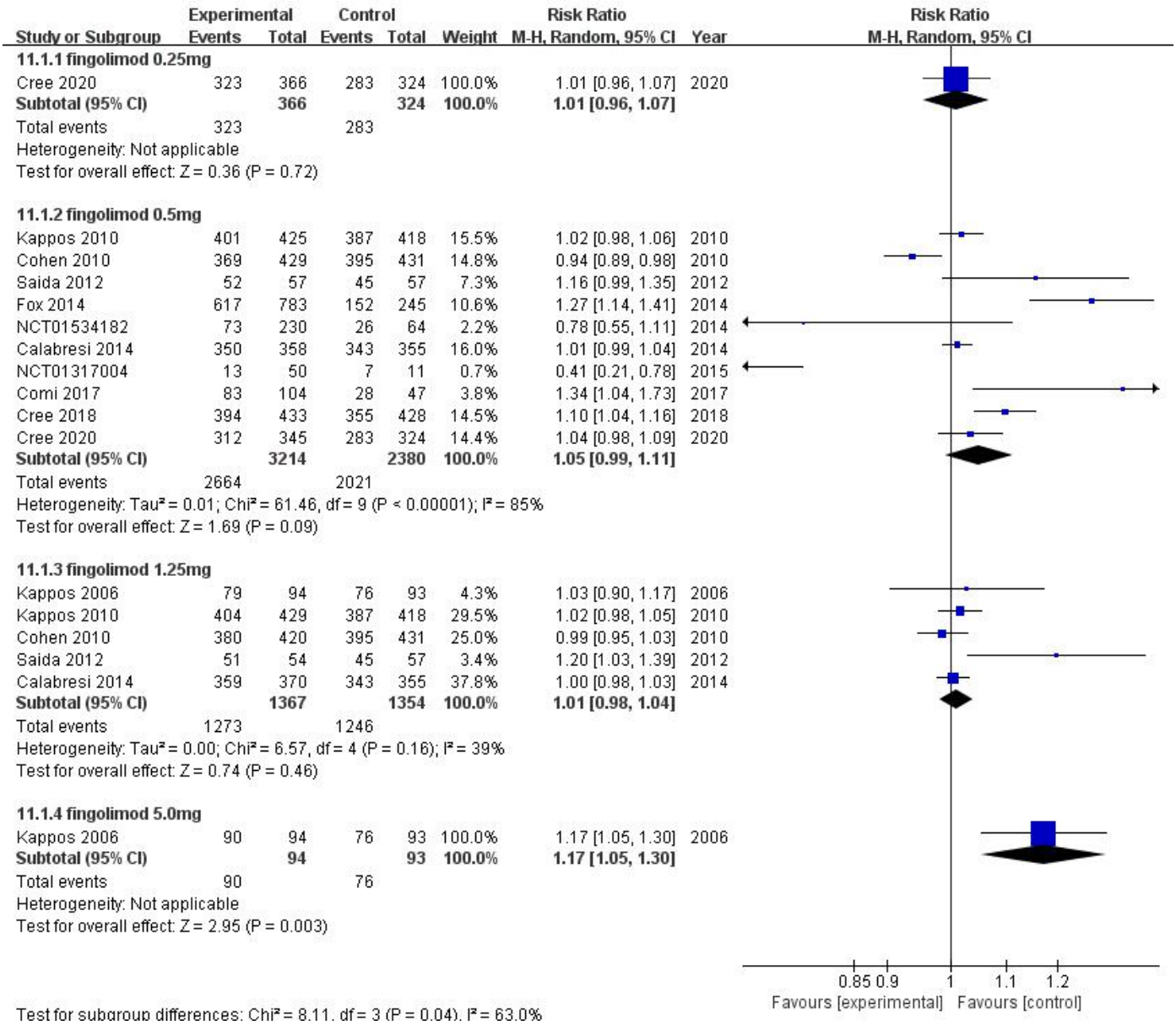

Experimental
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI Year
M-H, Random, 95% CI
11.1.1 fingolimod 0.25mg
Cree 2020 323 366 283 324 100.0% 1.01 [0.96, 1.07] 2020
Subtotal (95% CI) 366 324 100.0% 1.01 [0.96, 1.07]
Total events 323 283
Heterogeneity: Not applicable
Test for overall effect: Z = 0.36 (P = 0.72)
11.1.2 fingolimod 0.5mg
Kappos 2010 401 425 387 418 15.5% 1.02 [0.98, 1.06] 2010
Cohen 2010 369 429 395 431 14.8% 0.94 [0.89, 0.98] 2010
Saida 2012 52 57 45 57 7.3% 1.16 [0.99, 1.35] 2012
Fox 2014 617 783 152 245 10.6% 1.27 [1.14, 1.41] 2014
NCT01534182 73 230 26 64 2.2% 0.78 [0.55, 1.11] 2014
Calabresi 2014 350 358 343 355 16.0% 1.01 [0.99, 1.04] 2014
NCT01317004 13 50 7 11 0.7% 0.41 [0.21, 0.78] 2015
Comi 2017 83 104 28 47 3.8% 1.34 [1.04, 1.73] 2017
Cree 2018 394 433 355 428 14.5% 1.10 [1.04, 1.16] 2018
Cree 2020 312 345 283 324 14.4% 1.04 [0.98, 1.09] 2020
Subtotal (95% CI) 3214 2380 100.0% 1.05 [0.99, 1.11]
Total events 2664 2021
Heterogeneity: Tau² = 0.01; Chi² = 61.46, df = 9 (P < 0.00001); I² = 85%
Test for overall effect: Z = 1.69 (P = 0.09)
11.1.3 fingolimod 1.25mg
Kappos 2006 79 94 76 93 4.3% 1.03 [0.90, 1.17] 2006
Kappos 2010 404 429 387 418 29.5% 1.02 [0.98, 1.05] 2010
Cohen 2010 380 420 395 431 25.0% 0.99 [0.95, 1.03] 2010
Saida 2012 51 54 45 57 3.4% 1.20 [1.03, 1.39] 2012
Calabresi 2014 359 370 343 355 37.8% 1.00 [0.98, 1.03] 2014
Subtotal (95% CI) 1367 1354 100.0% 1.01 [0.98, 1.04]
Total events 1273 1246
Heterogeneity: Tau² = 0.00; Chi² = 6.57, df = 4 (P = 0.16); I² = 39%
Test for overall effect: Z = 0.74 (P = 0.46)
11.1.4 fingolimod 5.0mg
Kappos 2006 90 94 76 93 100.0% 1.17 [1.05, 1.30] 2006
Subtotal (95% CI) 94 93 100.0% 1.17 [1.05, 1.30]
Total events 90 76
Heterogeneity: Not applicable
Test for overall effect: Z = 2.95 (P = 0.003)
0.85 0.9 1 1.1 1.2
Favours [experimental] Favours [control]
Test for subgroup differences: Chi² = 8.11, df = 3 (P = 0.04), I² = 63.0%

Fig. S10 (SAEs)

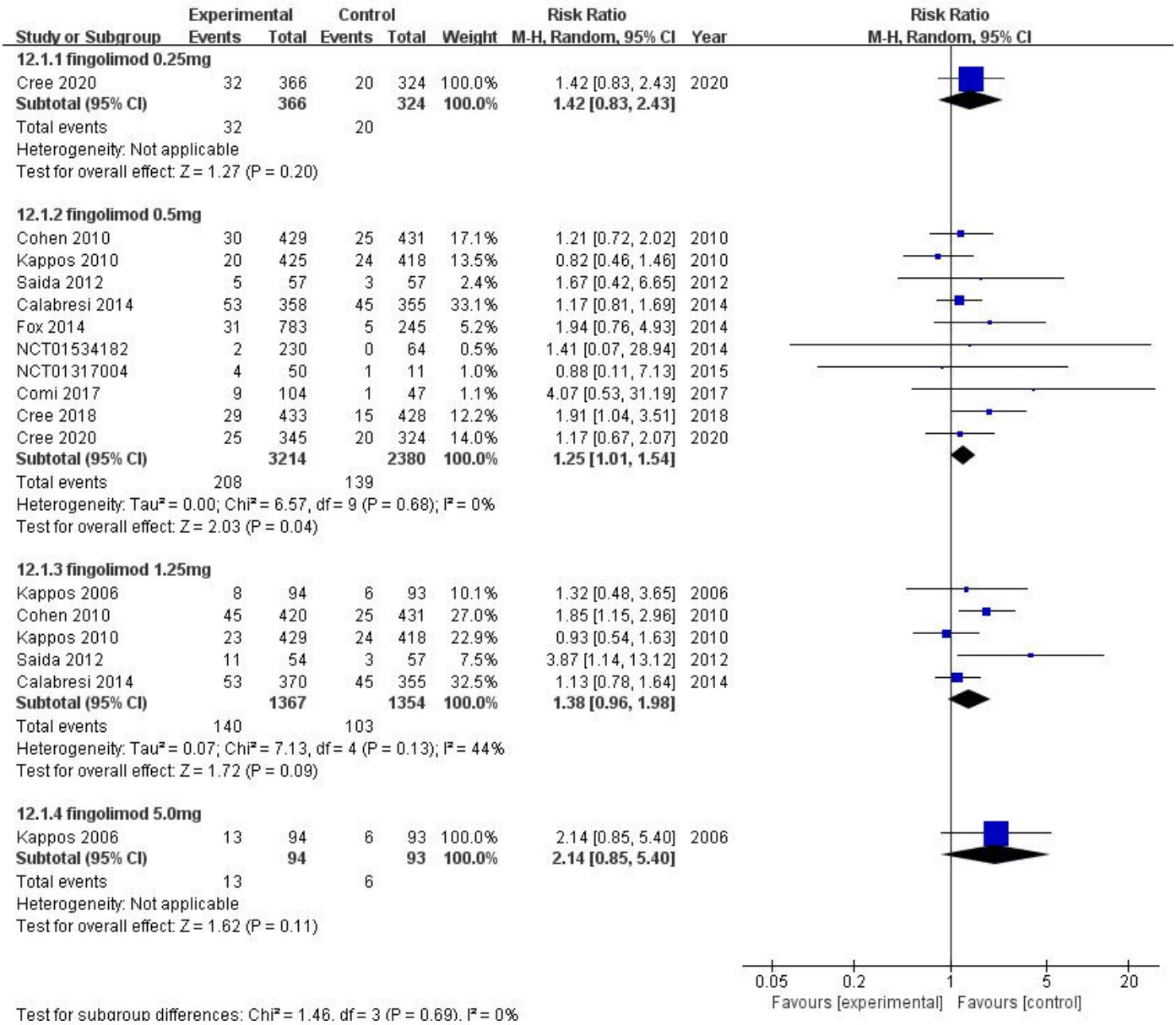
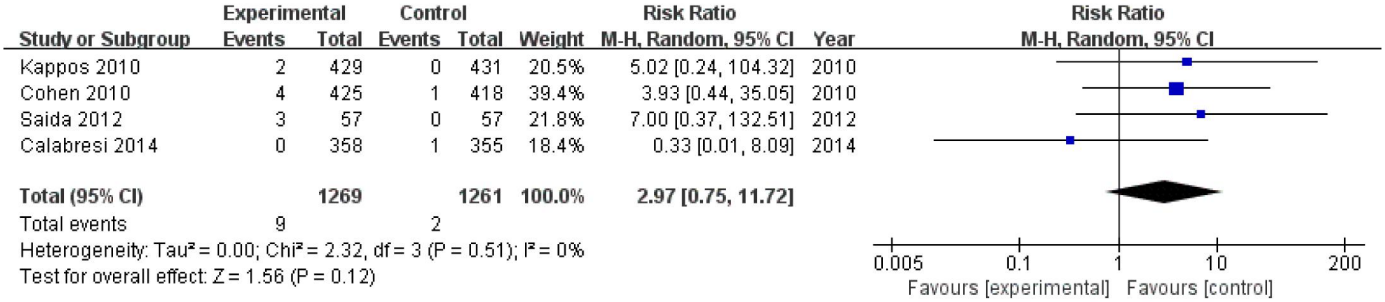
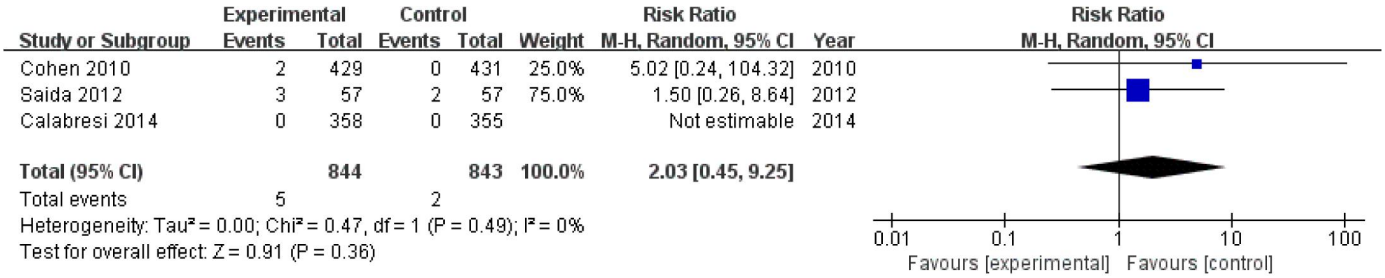

Experimental
Control
Risk Ratio
Risk Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI Year
M-H, Random, 95% CI
12.1.1 fingolimod 0.25mg
Cree 2020 32 366 20 324 100.0% 1.42 [0.83, 2.43] 2020
Subtotal (95% CI) 366 324 100.0% 1.42 [0.83, 2.43]
Total events 32 20
Heterogeneity: Not applicable
Test for overall effect: Z = 1.27 (P = 0.20)
12.1.2 fingolimod 0.5mg
Cohen 2010 30 429 25 431 17.1% 1.21 [0.72, 2.02] 2010
Kappos 2010 20 425 24 418 13.5% 0.82 [0.46, 1.46] 2010
Saida 2012 5 57 3 57 2.4% 1.67 [0.42, 6.65] 2012
Calabresi 2014 53 358 45 355 33.1% 1.17 [0.81, 1.69] 2014
Fox 2014 31 783 5 245 5.2% 1.94 [0.76, 4.93] 2014
NCT01534182 2 230 0 64 0.5% 1.41 [0.07, 28.94] 2014
NCT01317004 4 50 1 11 1.0% 0.88 [0.11, 7.13] 2015
Comi 2017 9 104 1 47 1.1% 4.07 [0.53, 31.19] 2017
Cree 2018 29 433 15 428 12.2% 1.91 [1.04, 3.51] 2018
Cree 2020 25 345 20 324 14.0% 1.17 [0.67, 2.07] 2020
Subtotal (95% CI) 3214 2380 100.0% 1.25 [1.01, 1.54]
Total events 208 139
Heterogeneity: Tau² = 0.00; Chi² = 6.57, df = 9 (P = 0.68); I² = 0%
Test for overall effect: Z = 2.03 (P = 0.04)
12.1.3 fingolimod 1.25mg
Kappos 2006 8 94 6 93 10.1% 1.32 [0.48, 3.65] 2006
Cohen 2010 45 420 25 431 27.0% 1.85 [1.15, 2.96] 2010
Kappos 2010 23 429 24 418 22.9% 0.93 [0.54, 1.63] 2010
Saida 2012 11 54 3 57 7.5% 3.87 [1.14, 13.12] 2012
Calabresi 2014 53 370 45 355 32.5% 1.13 [0.78, 1.64] 2014
Subtotal (95% CI) 1367 1354 100.0% 1.38 [0.96, 1.98]
Total events 140 103
Heterogeneity: Tau² = 0.07; Chi² = 7.13, df = 4 (P = 0.13); I² = 44%
Test for overall effect: Z = 1.72 (P = 0.09)
12.1.4 fingolimod 5.0mg
Kappos 2006 13 94 6 93 100.0% 2.14 [0.85, 5.40] 2006
Subtotal (95% CI) 94 93 100.0% 2.14 [0.85, 5.40]
Total events 13 6
Heterogeneity: Not applicable
Test for overall effect: Z = 1.62 (P = 0.11)
0.05
0.2
1
5
20
Favours [experimental]
Favours [control]
Test for subgroup differences: Chi² = 1.46, df = 3 (P = 0.69), I² = 0%

Fig. S11 (Sub-analysis of specific SAEs in the fingolimod 0.5 mg/d group)

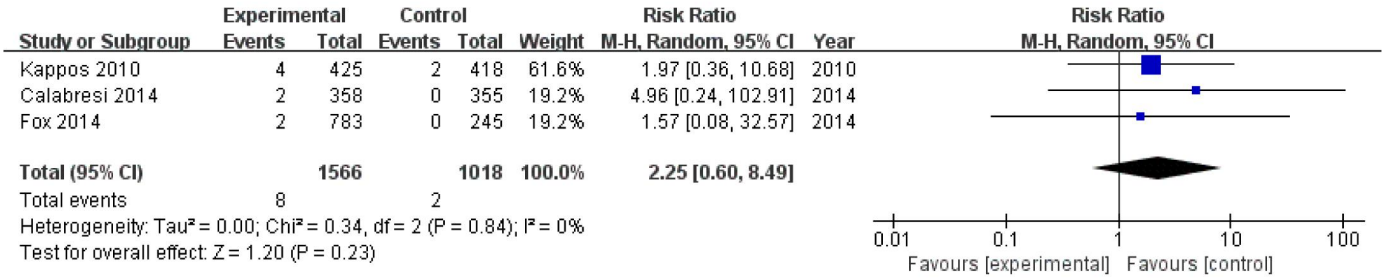
A. Bradycardia



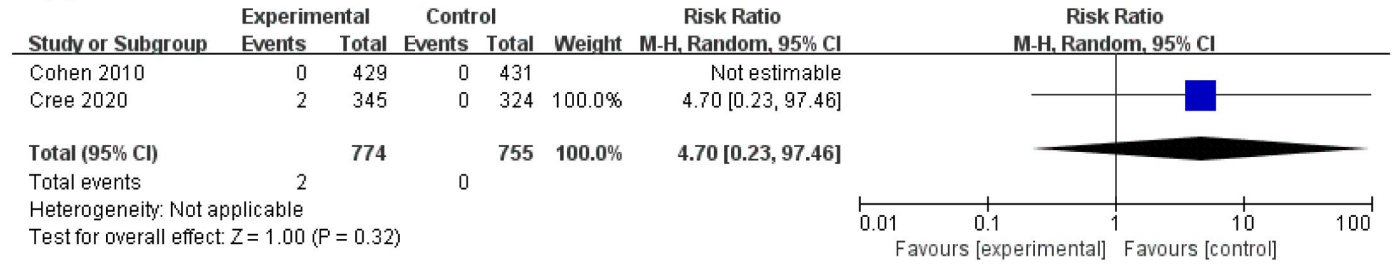
B. Atrioventricular block



C. Chest pain



D. Dyspnea



E. Infection

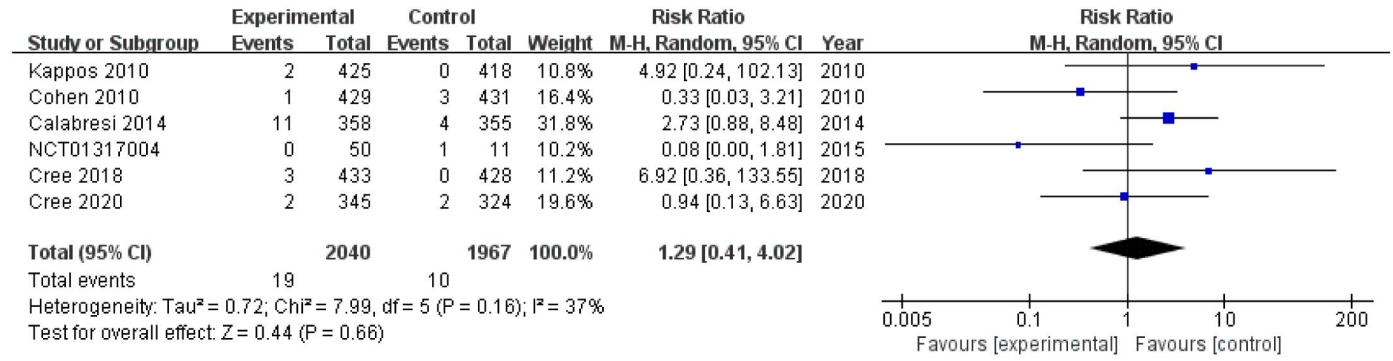
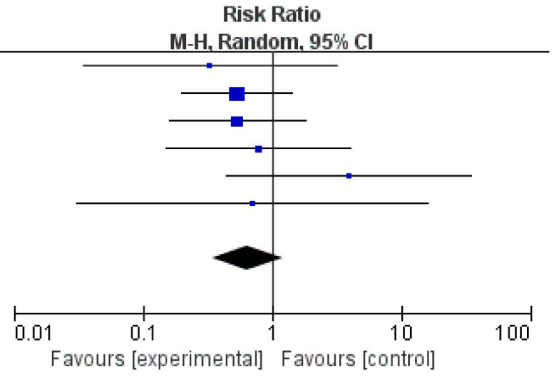


Fig. S12 (Sub-analysis of specific SAEs in the fingolimod 0.5 mg/d group)

A. MS relapse

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Calabresi 2014	1	358	3	355	7.6%	0.33 [0.03, 3.16]
Cree 2018	6	433	11	428	39.8%	0.54 [0.20, 1.44]
Cree 2020	4	345	7	324	26.0%	0.54 [0.16, 1.82]
Fox 2014	5	783	2	245	14.5%	0.78 [0.15, 4.01]
Kappos 2010	4	425	1	418	8.1%	3.93 [0.44, 35.05]
NCT01317004	1	50	0	11	3.9%	0.71 [0.03, 16.28]
Total (95% CI)		**2394**		**1781**	**100.0%**	**0.65 [0.35, 1.21]**
Total events	21		24			

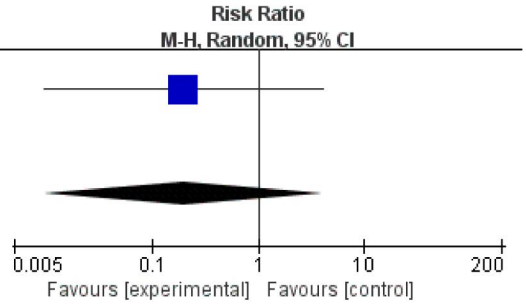
Heterogeneity: Tau² = 0.00; Chi² = 3.25, df = 5 (P = 0.66); I² = 0%
Test for overall effect: Z = 1.36 (P = 0.17)



B. Death

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Cohen 2010	0	429	0	431		Not estimable	2010
Kappos 2010	0	425	2	418	100.0%	0.20 [0.01, 4.09]	2010
Calabresi 2014	0	358	0	355		Not estimable	2014
Fox 2014	0	783	0	245		Not estimable	2014
Total (95% CI)		**1995**		**1449**	**100.0%**	**0.20 [0.01, 4.09]**	
Total events	0		2				

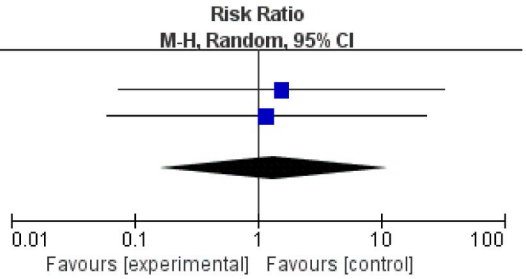
Heterogeneity: Not applicable
Test for overall effect: Z = 1.05 (P = 0.29)



C. Lymphopenia

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Kappos 2010	0	425	0	418		Not estimable	2010
Fox 2014	2	783	0	245	49.0%	1.57 [0.08, 32.57]	2014
NCT01317004	2	50	0	11	51.0%	1.18 [0.06, 22.94]	2015
Total (95% CI)		**1258**		**674**	**100.0%**	**1.35 [0.16, 11.31]**	
Total events	4		0				

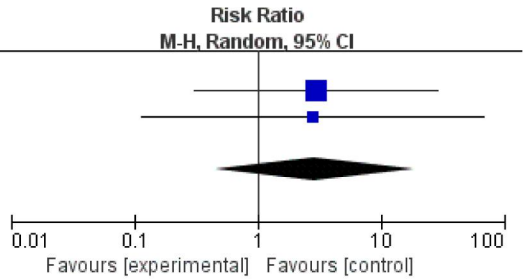
Heterogeneity: Tau² = 0.00; Chi² = 0.02, df = 1 (P = 0.89); I² = 0%
Test for overall effect: Z = 0.28 (P = 0.78)



D. Epilepsy

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Kappos 2010	0	425	0	418		Not estimable	2010
Calabresi 2014	3	358	1	355	66.7%	2.97 [0.31, 28.46]	2014
Cree 2020	1	345	0	324	33.3%	2.82 [0.12, 68.93]	2020
Total (95% CI)		**1128**		**1097**	**100.0%**	**2.92 [0.46, 18.48]**	
Total events	4		1				

Heterogeneity: Tau² = 0.00; Chi² = 0.00, df = 1 (P = 0.98); I² = 0%
Test for overall effect: Z = 1.14 (P = 0.25)



E. Depression

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Kappos 2010	0	425	1	418	27.9%	0.33 [0.01, 8.03]	2010
Calabresi 2014	1	358	5	355	41.6%	0.20 [0.02, 1.69]	2014
Cree 2020	3	433	0	428	30.5%	6.92 [0.36, 133.55]	2020
Total (95% CI)		**1216**		**1201**	**100.0%**	**0.67 [0.08, 6.03]**	
Total events	4		6				

Heterogeneity: Tau² = 1.81; Chi² = 3.86, df = 2 (P = 0.15); I² = 48%
Test for overall effect: Z = 0.35 (P = 0.72)

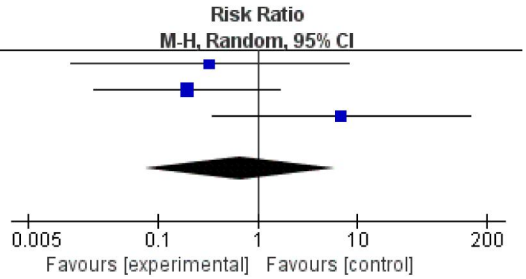
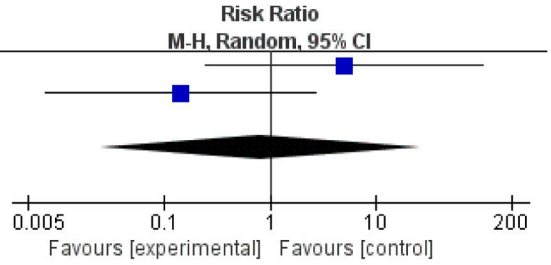


Fig. S13 (Sub-analysis of specific SAEs in the fingolimod 0.5 mg/d group)

A. Breast cancer

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Cohen 2010	2	429	0	431	49.6%	5.02 [0.24, 104.32]
Kappos 2010	0	425	3	418	50.4%	0.14 [0.01, 2.71]
Total (95% CI)		**854**		**849**	**100.0%**	**0.83 [0.02, 27.60]**
Total events	2		3			

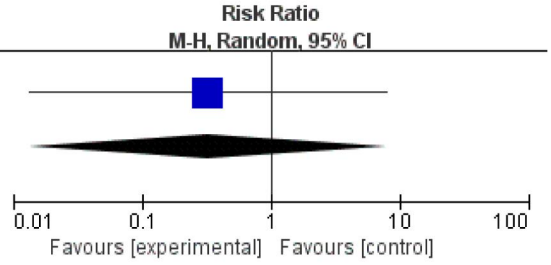
Heterogeneity: Tau² = 4.07; Chi² = 2.74, df = 1 (P = 0.10); I² = 64%
Test for overall effect: Z = 0.11 (P = 0.92)



B. Melanoma

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Cohen 2010	0	429	0	431		Not estimable
Kappos 2010	0	425	1	418	100.0%	0.33 [0.01, 8.03]
Total (95% CI)		**854**		**849**	**100.0%**	**0.33 [0.01, 8.03]**
Total events	0		1			

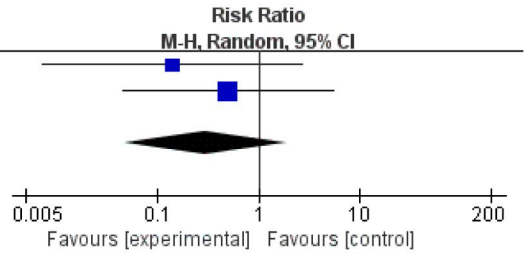
Heterogeneity: Not applicable
Test for overall effect: Z = 0.68 (P = 0.49)



C. Abortion

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Kappos 2010	0	425	3	418	39.6%	0.14 [0.01, 2.71]	2010
Calabresi 2014	1	358	2	355	60.4%	0.50 [0.05, 5.44]	2014
Total (95% CI)		**783**		**773**	**100.0%**	**0.30 [0.05, 1.94]**	
Total events	1		5				

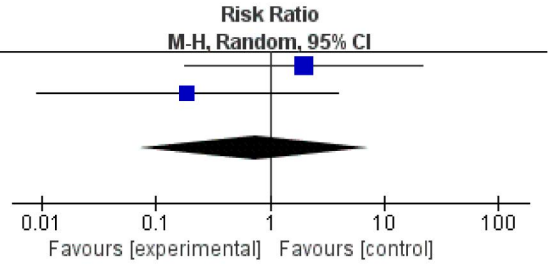
Heterogeneity: Tau² = 0.00; Chi² = 0.43, df = 1 (P = 0.51); I² = 0%
Test for overall effect: Z = 1.26 (P = 0.21)



D. Syncope

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Calabresi 2014	2	358	1	355	58.0%	1.98 [0.18, 21.77]
Cree 2020	0	345	2	324	42.0%	0.19 [0.01, 3.90]
Total (95% CI)		**703**		**679**	**100.0%**	**0.74 [0.07, 7.34]**
Total events	2		3			

Heterogeneity: Tau² = 0.88; Chi² = 1.45, df = 1 (P = 0.23); I² = 31%
Test for overall effect: Z = 0.26 (P = 0.79)



E. Macular edema

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI	Year
Kappos 2010	0	425	0	418		Not estimable	2010
Calabresi 2014	1	358	1	355	100.0%	0.99 [0.06, 15.79]	2014
Total (95% CI)		**783**		**773**	**100.0%**	**0.99 [0.06, 15.79]**	
Total events	1		1				

Heterogeneity: Not applicable
Test for overall effect: Z = 0.01 (P = 1.00)

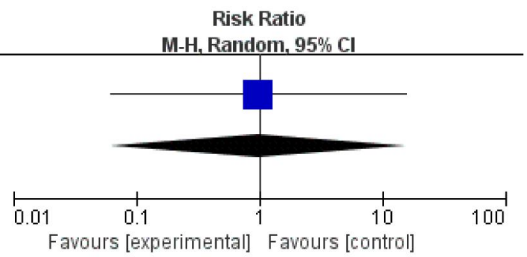


Fig. S14 (Sub-analysis of basal-cell carcinoma in the fingolimod 0.5 mg/d group)

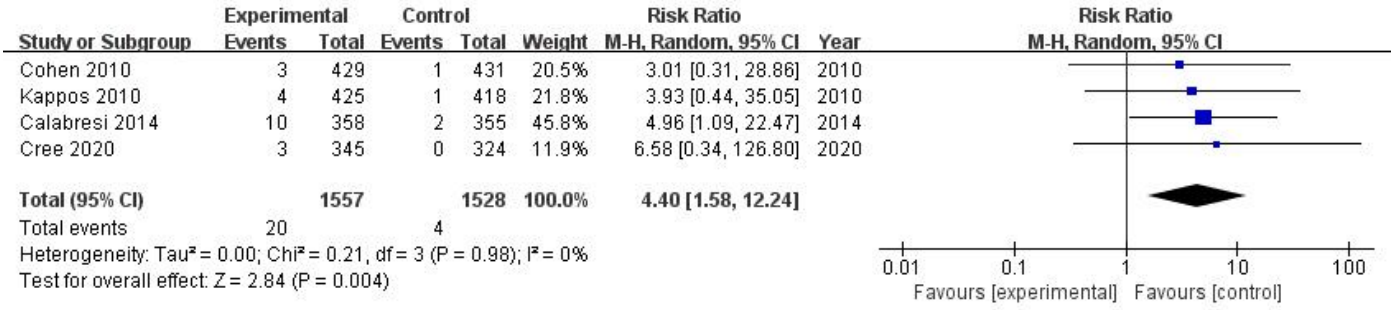
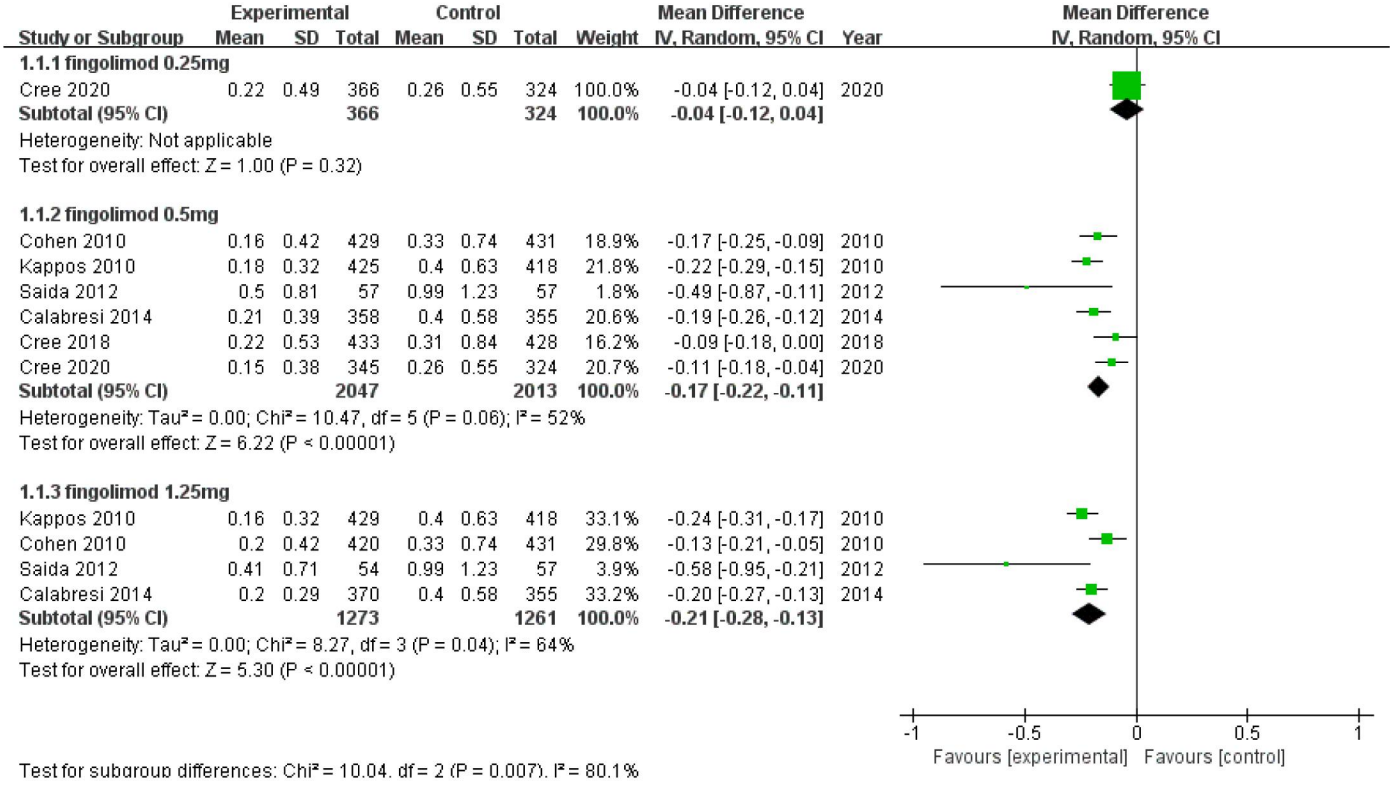


Fig. S15

A


Experimental
Control
Mean Difference
Mean Difference
Study or Subgroup Mean SD Total Mean SD Total Weight IV, Random, 95% CI Year
IV, Random, 95% CI
1.1.1 fingolimod 0.25mg
Cree 2020 0.22 0.49 366 0.26 0.55 324 100.0% -0.04 [-0.12, 0.04] 2020
Subtotal (95% CI) 366 324 100.0% -0.04 [-0.12, 0.04]
Heterogeneity: Not applicable
Test for overall effect: Z = 1.00 (P = 0.32)
1.1.2 fingolimod 0.5mg
Cohen 2010 0.16 0.42 429 0.33 0.74 431 18.9% -0.17 [-0.25, -0.09] 2010
Kappos 2010 0.18 0.32 425 0.4 0.63 418 21.8% -0.22 [-0.29, -0.15] 2010
Saida 2012 0.5 0.81 57 0.99 1.23 57 1.8% -0.49 [-0.87, -0.11] 2012
Calabresi 2014 0.21 0.39 358 0.4 0.58 355 20.6% -0.19 [-0.26, -0.12] 2014
Cree 2018 0.22 0.53 433 0.31 0.84 428 16.2% -0.09 [-0.18, 0.00] 2018
Cree 2020 0.15 0.38 345 0.26 0.55 324 20.7% -0.11 [-0.18, -0.04] 2020
Subtotal (95% CI) 2047 2013 100.0% -0.17 [-0.22, -0.11]
Heterogeneity: Tau² = 0.00; Chi² = 10.47, df = 5 (P = 0.06); I² = 52%
Test for overall effect: Z = 6.22 (P < 0.00001)
1.1.3 fingolimod 1.25mg
Kappos 2010 0.16 0.32 429 0.4 0.63 418 33.1% -0.24 [-0.31, -0.17] 2010
Cohen 2010 0.2 0.42 420 0.33 0.74 431 29.8% -0.13 [-0.21, -0.05] 2010
Saida 2012 0.41 0.71 54 0.99 1.23 57 3.9% -0.58 [-0.95, -0.21] 2012
Calabresi 2014 0.2 0.29 370 0.4 0.58 355 33.2% -0.20 [-0.27, -0.13] 2014
Subtotal (95% CI) 1273 1261 100.0% -0.21 [-0.28, -0.13]
Heterogeneity: Tau² = 0.00; Chi² = 8.27, df = 3 (P = 0.04); I² = 64%
Test for overall effect: Z = 5.30 (P < 0.00001)
Test for subgroup differences: Chi² = 10.04, df = 2 (P = 0.007), I² = 80.1%
-1
-0.5
0
0.5
1
Favours [experimental]
Favours [control]

B

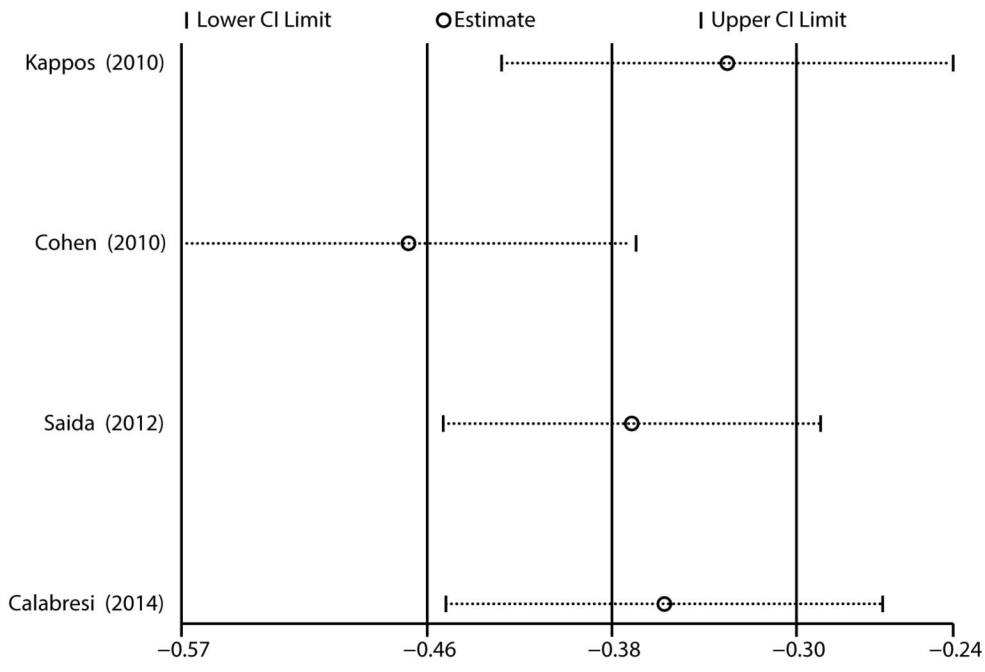

Lower CI Limit
Estimate
Upper CI Limit
Kappos (2010)
Cohen (2010)
Saida (2012)
Calabresi (2014)
−0.57
−0.46
−0.38
−0.30
−0.24

Fig. S16

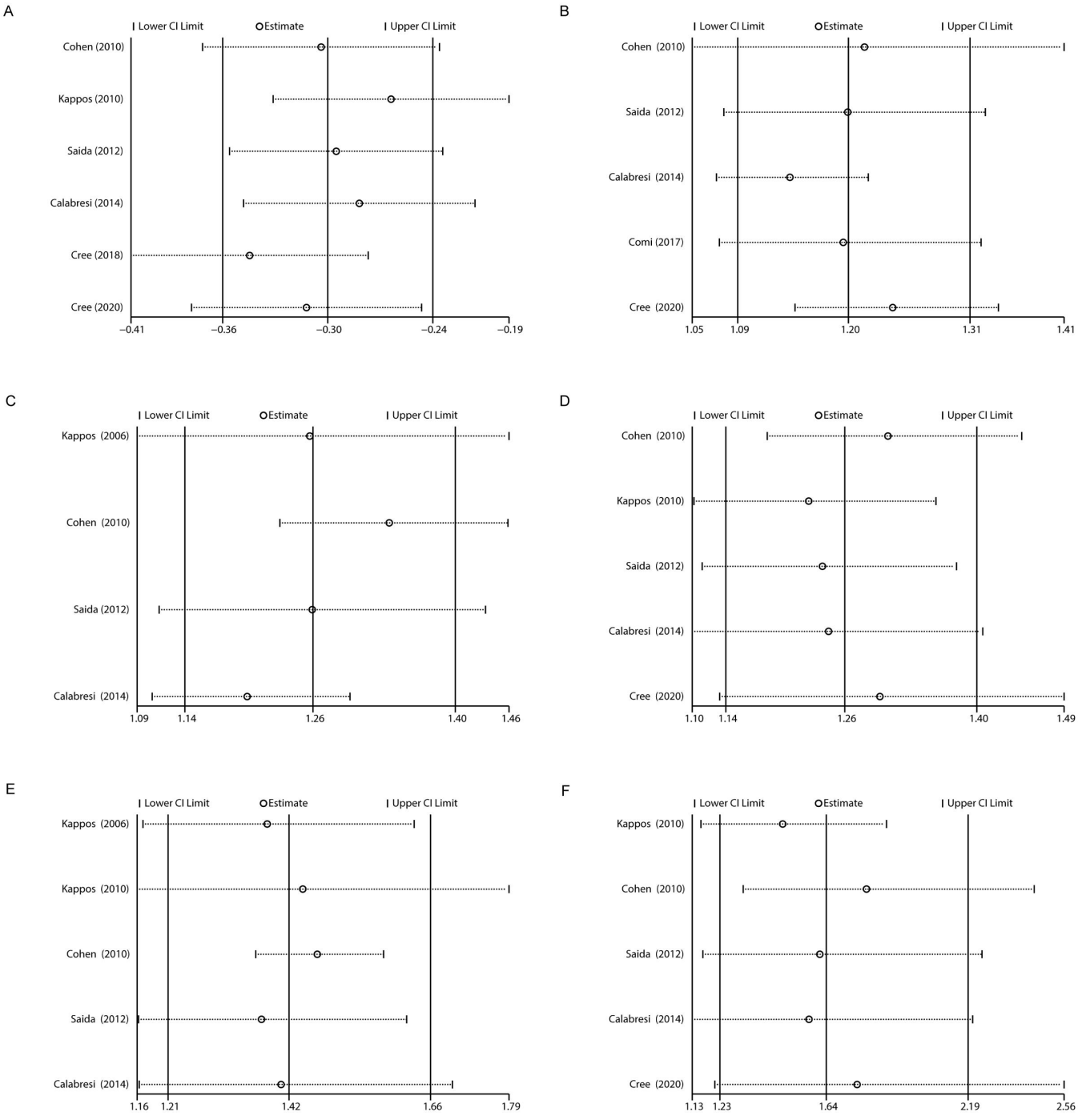
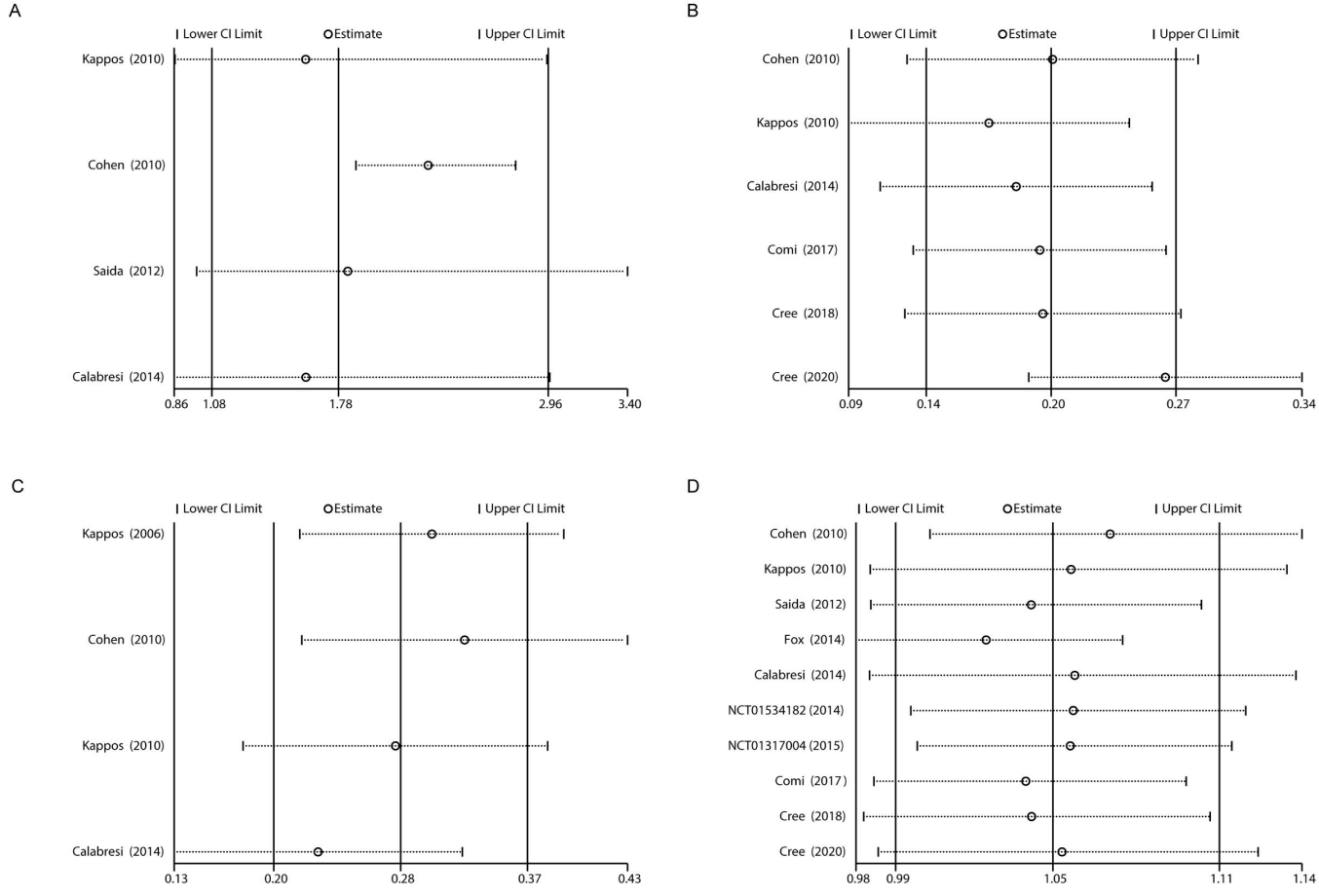

A
Lower CI Limit
Estimate
Upper CI Limit
Cohen (2010)
Kappos (2010)
Saida (2012)
Calabresi (2014)
Cree (2018)
Cree (2020)
−0.41
−0.36
−0.30
−0.24
−0.19
B
Lower CI Limit
Estimate
Upper CI Limit
Cohen (2010)
Saida (2012)
Calabresi (2014)
Comi (2017)
Cree (2020)
1.05
1.09
1.20
1.31
1.41
C
Lower CI Limit
Estimate
Upper CI Limit
Kappos (2006)
Cohen (2010)
Saida (2012)
Calabresi (2014)
1.09
1.14
1.26
1.40
1.46
D
Lower CI Limit
Estimate
Upper CI Limit
Cohen (2010)
Kappos (2010)
Saida (2012)
Calabresi (2014)
Cree (2020)
1.10
1.14
1.26
1.40
1.49
E
Lower CI Limit
Estimate
Upper CI Limit
Kappos (2006)
Kappos (2010)
Cohen (2010)
Saida (2012)
Calabresi (2014)
1.16
1.21
1.42
1.66
1.79
F
Lower CI Limit
Estimate
Upper CI Limit
Kappos (2010)
Cohen (2010)
Saida (2012)
Calabresi (2014)
Cree (2020)
1.13
1.23
1.64
2.19
2.56

Fig. S17


A
Lower CI Limit
Estimate
Upper CI Limit
Kappos (2010)
Cohen (2010)
Saida (2012)
Calabresi (2014)
0.86
1.08
1.78
2.96
3.40
B
Lower CI Limit
Estimate
Upper CI Limit
Cohen (2010)
Kappos (2010)
Calabresi (2014)
Comi (2017)
Cree (2018)
Cree (2020)
0.09
0.14
0.20
0.27
0.34
C
Lower CI Limit
Estimate
Upper CI Limit
Kappos (2006)
Cohen (2010)
Kappos (2010)
Calabresi (2014)
0.13
0.20
0.28
0.37
0.43
D
Lower CI Limit
Estimate
Upper CI Limit
Cohen (2010)
Kappos (2010)
Saida (2012)
Fox (2014)
Calabresi (2014)
NCT01534182 (2014)
NCT01317004 (2015)
Comi (2017)
Cree (2018)
Cree (2020)
0.98
0.99
1.05
1.11
1.14